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CS 2300 : Discrete Computational Structures
Summer Semester, 2025
ASSIGNMENT #6
Due Date: Saturday, July 5

These are the problems that you need to turn in. Always explain your answers and show your reasoning. **You will be graded based on how well you explain your answers. Just correct answers will not be enough!**

The total points add up to 100, but you will be graded out of 80. This means you have the opportunity to earn 20 extra credit points.

1. [12 Pts] 12 women and 10 men are on the faculty. How many ways are there to pick a committee of 6 if (a) Ann and Beth will not serve together, (b) at least one woman must be chosen, (c) at least one man and one woman must be chosen.

(a) Ann and Beth will not serve together

Total number of committees: $\binom{22}{6}$

Number of committees with both Ann and Beth: $\binom{20}{4}$

Valid committees (Ann and Beth not both on the committee): $\binom{22}{6} - \binom{20}{4} = 74613 - 4845 = 69768$

(b) At least one woman must be chosen

Total number of committees: $\binom{22}{6}$

Committees with no women (all men): $\binom{10}{6}$

Valid committees (at least one woman):

$$\binom{22}{6} - \binom{10}{6} = 74613 - 210 = 74403$$

(c) At least one man and one woman must be chosen

Total number of committees: $\binom{22}{6}$

Committees with only women: $\binom{12}{6}$

Committees with only men: $\binom{10}{6}$

Valid committees (at least one man and one woman): $\binom{22}{6} - \binom{12}{6} - \binom{10}{6} = 74613 - 924 - 210 = 73479$

2. [8 Pts] Prove, using a combinatorial argument, that $C(m+n, 2) = C(m, 2) + C(n, 2) + mn$, where $m, n \geq 2$.

I have a group of $m+n$ people.

Divided into two subgroups:

Group A: m people

Group B: n people

I want to count the number of ways to choose 2 people from the total group of $m+n$ people.

Left-Hand Side:

$$\binom{m+n}{2}$$

This is the total number of ways to choose any 2 people from the entire group of $m+n$.

Right-Hand Side:

Count based on the composition of the pair:

Both people from Group A: $\binom{m}{2}$ ways

Both people from Group B: $\binom{n}{2}$ ways

One person from Group A and one from Group B: $m \cdot n$ ways

Adding these cases, I get: $\binom{m}{2} + \binom{n}{2} + mn$

Both sides count the number of ways to choose 2 people from a group of $m+n$, but in different ways.

Thus, they must be equal:

$$\binom{m+n}{2} = \binom{m}{2} + \binom{n}{2} + mn$$

3. [10 Pts] Prove, (a) using a combinatorial argument, and (b) using an algebraic proof, that $P(n, 3)C(n - 3, k - 3) = C(n, k)P(k, 3)$.

(a) Combinatorial Argument

I can interpret both sides as counting the same thing in two different ways.

Left-Hand Side: $P(n, 3) \cdot \binom{n-3}{k-3}$

First, I can choose and arrange 3 elements from an n -element set. This can be done in $P(n, 3)$ ways.

Then, I choose $k - 3$ additional elements from the remaining $n - 3$ elements. This can be done in $\binom{n-3}{k-3}$ ways.

So the left-hand side counts the number of ways to choose and order 3 elements from a set of n , and then choose $k - 3$ more to form a k -element subset.

Right-hand side: $\binom{n}{k} \cdot P(k, 3)$

First, I can choose a k -element subset from the n -element set. This can be done in $\binom{n}{k}$ ways.

Then, I choose and order 3 elements from the k -element subset. This can be done in $P(k, 3)$ ways.

So the right-hand side also counts the number of ways to form a k -element subset and then order 3 elements from within it.

Since both sides count the same thing, the identity holds combinatorially.

(b) Algebraic Proof

$$P(n, 3) \cdot \binom{n-3}{k-3} = \frac{n!}{(n-3)!} \cdot \frac{(n-3)!}{(k-3)!(n-k)!} = \frac{n!}{(k-3)!(n-k)!}$$

$$\binom{n}{k} \cdot P(k, 3) = \frac{n!}{k!(n-k)!} \cdot \frac{k!}{(k-3)!} = \frac{n!}{(n-k)!} \cdot \frac{1}{(k-3)!} = \frac{n!}{(k-3)!(n-k)!}$$

Therefore, both sides are equal algebraically, and the identity is proven.

4. [12 Pts] A cookie shop sells 4 different kinds of cookies. How many different ways are there to choose 10 cookies if (a) you have no restrictions? (b) you pick at least two of each? (c) you pick at least 3 oatmeal cookies and at most 3 sugar cookies?

Let the 4 kinds be:

a = chocolate chip

b = oatmeal

c = peanut butter

d = sugar

(a) No restrictions

I am solving for the number of non-negative integer solutions to: $a+b+c+d = 10$, where $a, b, c, d \geq 0$

Using the stars and bars formula: $\binom{10+4-1}{4-1} = \binom{13}{3} = 286$

286

(b) At least two of each kind

I now require: $a, b, c, d \geq 2$

Let $a' = a - 2$, $b' = b - 2$, $c' = c - 2$, $d' = d - 2$.

Then I solve: $a' + b' + c' + d' = 10 - 4 \times 2 = 2$

Number of non-negative integer solutions: $\binom{2+4-1}{4-1} = \binom{5}{3} = 10$

10

(c) At least 3 oatmeal and at most 3 sugar cookies

I need: $b \geq 3$, $d \leq 3$, $a + b + c + d = 10$

Let $b' = b - 3$ so that $b' \geq 0$.

Then: $a + b' + 3 + c + d = 10 \Rightarrow a + b' + c + d = 7$, with $d \leq 3$

First, I count all non-negative integer solutions to: $a + b' + c + d = 7$

Total: $\binom{7+4-1}{4-1} = \binom{10}{3} = 120$

Next, I subtract the cases where $d > 3$.

Let $d' = d - 4$, then $d' \geq 0$: $a + b' + c + d' = 3 \Rightarrow \binom{6}{3} = 20$

So the number of valid combinations is: $120 - 20 = 100$

100

5. [4 Pts] If I have 5 bananas, 3 oranges, and 4 apples, how many ways can I distribute these to 11 friends, if each friend gets one fruit (so one fruit is left over)? Describe the kind of combinatorial problem you are solving.

I am given 12 fruits:

5 bananas

3 oranges

4 apples

I am distributing these fruits to 11 friends, with each friend getting exactly one fruit. That means 1 fruit is left over.

The fruits are of three types (bananas, oranges, apples) and are indistinguishable within each type. The friends are distinguishable.

I need to choose which fruit is left out

There are 12 possible fruits that can be left out:

$$\binom{5}{1} + \binom{3}{1} + \binom{4}{1} = 5 + 3 + 4 = 12 \text{ choices}$$

I break the counting into three cases based on the type of fruit removed.

Case 1: One banana is left out

Remaining fruits: 4 bananas, 3 oranges, 4 apples.

$$\text{Number of distributions: } 5 \cdot \frac{11!}{4! \cdot 3! \cdot 4!} = 5 \cdot \frac{39916800}{24 \cdot 6 \cdot 24} = 5 \cdot \frac{39916800}{3456} = 5 \cdot 11552 = 57760$$

Case 2: One orange is left out

Remaining fruits: 5 bananas, 2 oranges, 4 apples.

$$\text{Number of distributions: } 3 \cdot \frac{11!}{5! \cdot 2! \cdot 4!} = 3 \cdot \frac{39916800}{120 \cdot 2 \cdot 24} = 3 \cdot \frac{39916800}{5760} = 3 \cdot 6934 = 20802$$

Case 3: One apple is left out

Remaining fruits: 5 bananas, 3 oranges, 3 apples.

$$\text{Number of distributions: } 4 \cdot \frac{11!}{5! \cdot 3! \cdot 3!} = 4 \cdot \frac{39916800}{120 \cdot 6 \cdot 6} = 4 \cdot \frac{39916800}{4320} = 4 \cdot 9240 = 36960$$

$$\text{Total number of ways: } 57760 + 20802 + 36960 = 115522$$

6. [12 Pts] How many ways are there to pack 24 different books into 6 boxes with 4 books each if (a) all 6 boxes are sent to different addresses? (b) all 6 boxes are sent to the same address? (c) 3 of the boxes are shipped to three different addresses while 3 are left to be addressed later?

(a) All 6 boxes are sent to different addresses?

Since all boxes are distinguishable, I count the number of ways to divide the 24 distinct books into 6 ordered groups of 4:

$$\binom{24}{4} \cdot \binom{20}{4} \cdot \binom{16}{4} \cdot \binom{12}{4} \cdot \binom{8}{4} \cdot \binom{4}{4}$$

(b) All 6 boxes are sent to the same address?

Since all boxes are indistinguishable, I divide by the number of permutations of the 6 groups (i.e., $6!$):

$$\frac{1}{6!} \cdot \binom{24}{4} \cdot \binom{20}{4} \cdot \binom{16}{4} \cdot \binom{12}{4} \cdot \binom{8}{4} \cdot \binom{4}{4}$$

(c) 3 of the boxes are shipped to three different addresses, while 3 are left to be addressed later?

In this case, 3 boxes are distinguishable (addressed) and 3 are indistinguishable (unlabeled). First, I divide the books into 6 unordered groups of 4, then choose 3 of them to label and assign the addresses:

$$\frac{1}{3!} \cdot \binom{24}{4} \cdot \binom{20}{4} \cdot \binom{16}{4} \cdot \binom{12}{4} \cdot \binom{8}{4} \cdot \binom{4}{4} \cdot \binom{6}{3} \cdot 3!$$

Simplifying (since $\frac{1}{3!} \cdot 3! = 1$), this becomes:

$$\binom{6}{3} \cdot \binom{24}{4} \cdot \binom{20}{4} \cdot \binom{16}{4} \cdot \binom{12}{4} \cdot \binom{8}{4} \cdot \binom{4}{4}$$

7. [10 Pts] (a) How many ways are there to pack 5 identical books into 5 identical boxes with no restrictions placed on how many can go in a box (some boxes can be empty)? (b) What if the books are different?

(a) How many ways are there to pack 5 identical books into 5 identical boxes with no restrictions placed on how many can go in a box (some boxes can be empty)?

Since everything is identical, I only need to care about the number of group sizes, not which box gets which.

This is the same as counting the number of ways to partition the number 5, which is:

Answer: 7 ways

The partitions are:

5

4 + 1

3 + 2

3 + 1 + 1

2 + 2 + 1

2 + 1 + 1 + 1

1 + 1 + 1 + 1 + 1

(b) What if the books are different?

Now the books are different, but the boxes are still identical.

This asks how many ways can I split 5 distinct books into non-empty groups, where the order of groups doesn't matter?

This is given by the Bell number, and:

Answer: 52 ways

8. [10 Pts] How many ways can we place 8 books on a bookcase with 5 shelves if the books are (a) indistinguishable copies (b) all distinct? Note that the position of the books on the shelves matter. Note that all books may be placed on the same shelf.

(a) Indistinguishable books:

I am distributing 8 identical books to 5 distinct shelves, where each shelf can hold any number of books (including zero). For this I have to use stars and bars.

$$\text{Number of ways} = \binom{8+5-1}{5-1} = \binom{12}{4} = 495$$

495

(b) Distinct books:

Each book is distinct, and both the shelf and the position on the shelf matter.

I count the number of ways to:

Assign each book to a shelf: 5^8 possibilities,

Arrange the books on the shelves.

$$\text{Since } \binom{8}{n_1, n_2, n_3, n_4, n_5} = \frac{8!}{n_1! \cdot n_2! \cdot n_3! \cdot n_4! \cdot n_5!},$$

multiplying by the product of the factorials gives: $\frac{8!}{n_1! \cdots n_5!} \cdot (n_1! \cdots n_5!) = 8!$

So each distribution contributes exactly $8!$ arrangements.

There are $\binom{12}{4} = 495$ such distributions.

$$\text{Total number of ways} = 495 \cdot 8! = 495 \cdot 40320 = 19958400$$

9. [4 Pts] Consider a simple, undirected graph G that has 6 vertices with degrees of 5, 4, 3, 3, 1, 1. Can G exist? Explain.

In any simple undirected graph, the sum of the degrees of all vertices equals twice the number of edges:

$$\sum \text{degrees of all vertices} = 2 \times \text{number of edges}$$

Therefore, the sum of the degrees must be even.

Compute:

$$5 + 4 + 3 + 3 + 1 + 1 = 17$$

Since 17 is odd, this violates the Handshaking Lemma.

A graph with degrees 5, 4, 3, 3, 1, 1 cannot exist because the sum of degrees is odd. Thus, the sequence is not graphical.

10. [6 Pts] If G is a simple, undirected graph with n vertices and $n - 1$ edges, (a) is G connected? (b) is G acyclic? For each question, if *yes*, give a short justification. If *no*, give a counterexample.

(a) Is G connected?

Answer: No

A graph with n vertices and $n - 1$ edges may or may not be connected.

Counterexample:

Let G have 4 vertices: $\{v_1, v_2, v_3, v_4\}$, and suppose the edges are $\{v_1, v_2\}$, $\{v_3, v_4\}$, and $\{v_1, v_3\}$.

This graph has 3 edges (which is $n - 1$) but is disconnected because v_4 is isolated from the rest of the graph.

Therefore, a graph with $n - 1$ edges is not connected.

(b) Is G acyclic?

Answer: Yes

Any simple undirected graph with n vertices and $n - 1$ edges must be acyclic.

A cycle in a simple graph requires at least as many edges as vertices.

If G contained a cycle, it would require at least n edges.

Since G has only $n - 1$ edges, it cannot contain a cycle.

Thus, G is acyclic.

11. [12 Pts] Prove that a simple, undirected graph is a tree if and only if it is acyclic but adding any edge will create a cycle.

Proof: I need to prove both directions of the equivalence.

Right:

Suppose G is a tree

Then:

G is connected and acyclic by definition of a tree.

For any pair of vertices u and v , there is a unique path between them (since trees are minimally connected).

If I add an edge (u, v) , I create a cycle consisting of the new edge and the existing path from u to v .

Therefore, adding any edge to G creates a cycle.

Left:

Suppose G is acyclic and adding any edge creates a cycle.

I want to show that G is connected:

Suppose for contradiction that G is not connected.

Then there exist vertices u and v in different connected components.

Since there is no path between u and v , adding the edge (u, v) cannot form a cycle.

This contradicts the assumption that adding any edge creates a cycle.

Hence, G must be connected.

Since G is both connected and acyclic, it is a tree.